
title: "GDP per Capita and Child Mortality (2020)" author: "Analysis Week 2" date: "May 19, 2026" output:
github_document: toc: true toc_depth: 2 html_document: self_contained: true

Introduction

This report compares **GDP per capita** and **under-five child mortality** (deaths per 1,000 live births) across countries in **2020**, using the two datasets in this folder. Values are taken from the `2020` column in each wide-format file (`geo` , `name` , and `yearly` columns).

Data preparation

A quick look at the 2020 subsets:

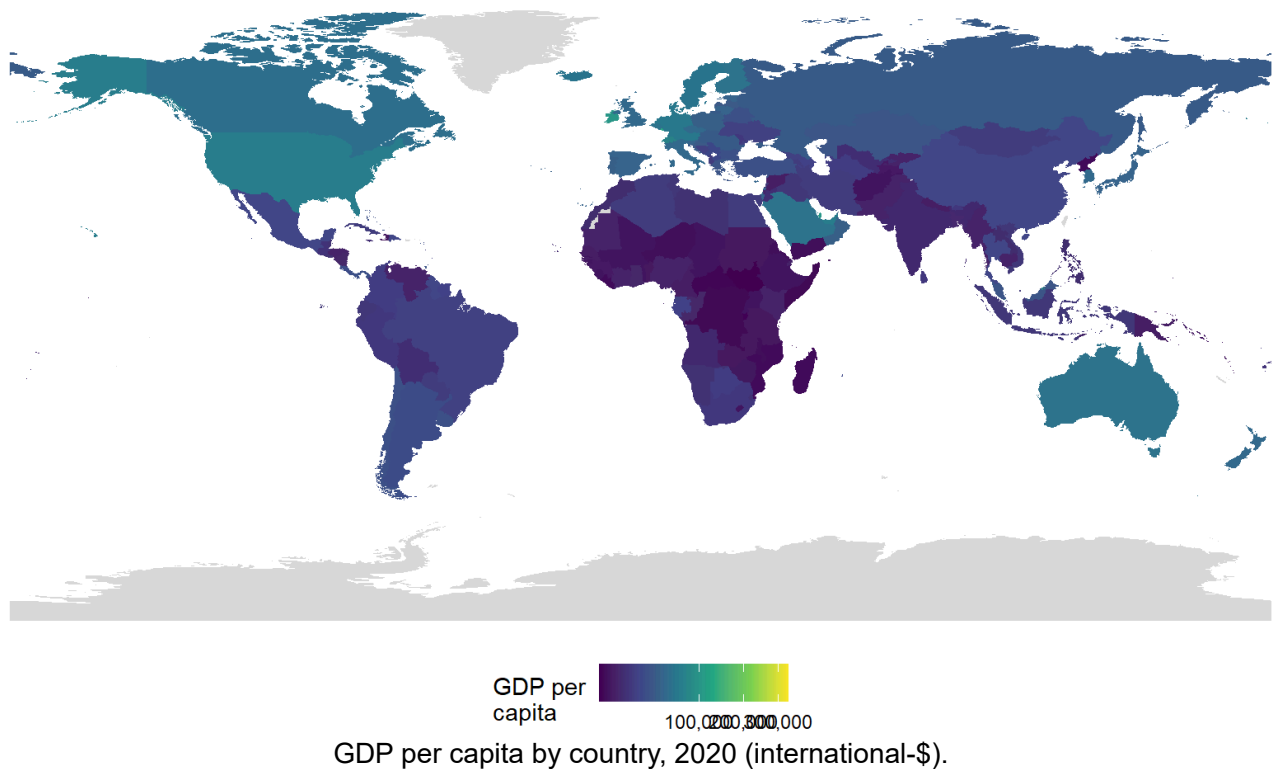
ISO3	Country	GDP per capita (2020)
MCO	Monaco	332202
LUX	Luxembourg	129866
SGP	Singapore	115893
QAT	Qatar	103062
IRL	Ireland	101971

ISO3	Country	Child mortality (2020)
SOM	Somalia	114.4
NGA	Nigeria	114.1
TCD	Chad	110.0
SLE	Sierra Leone	107.8
CAF	Central African Republic	102.9

In **193** countries GDP per capita is available for 2020; **194** countries have child mortality data. **193** countries appear in both datasets and are used in the scatter plot.

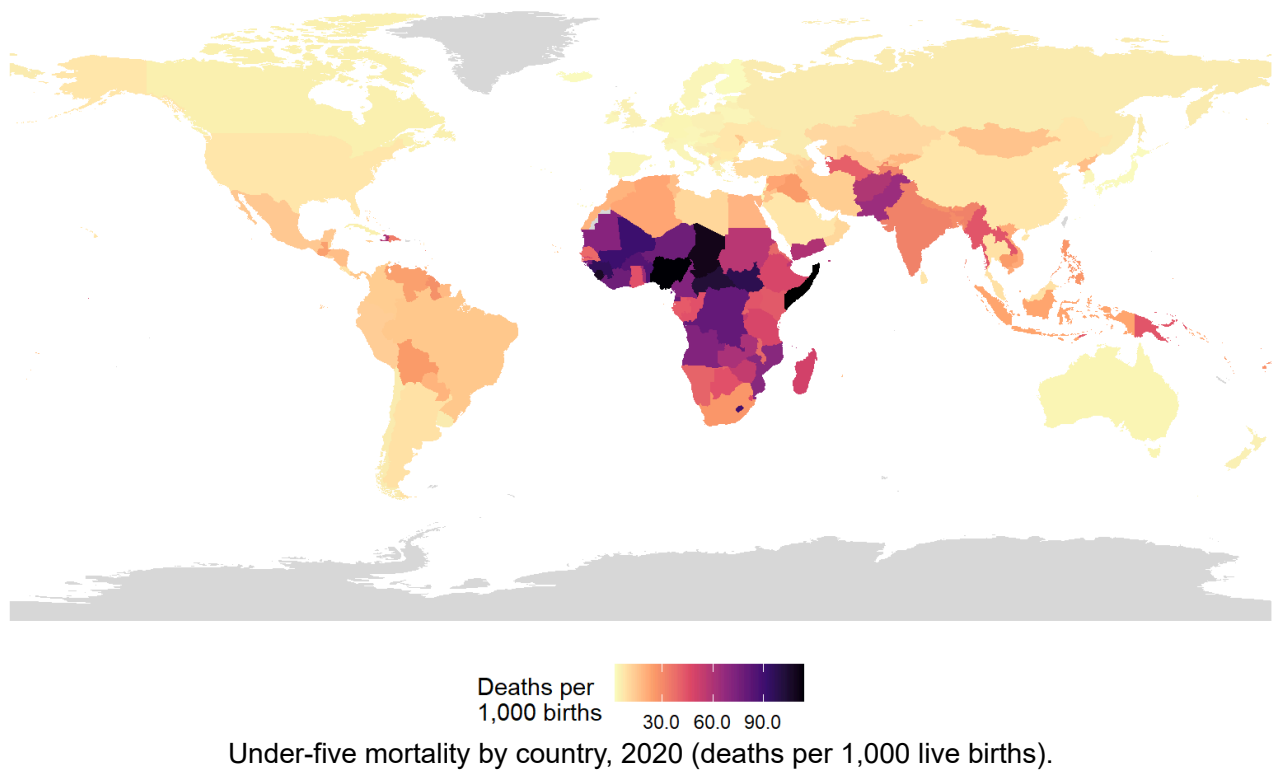
GDP per capita in 2020

GDP per capita, 2020

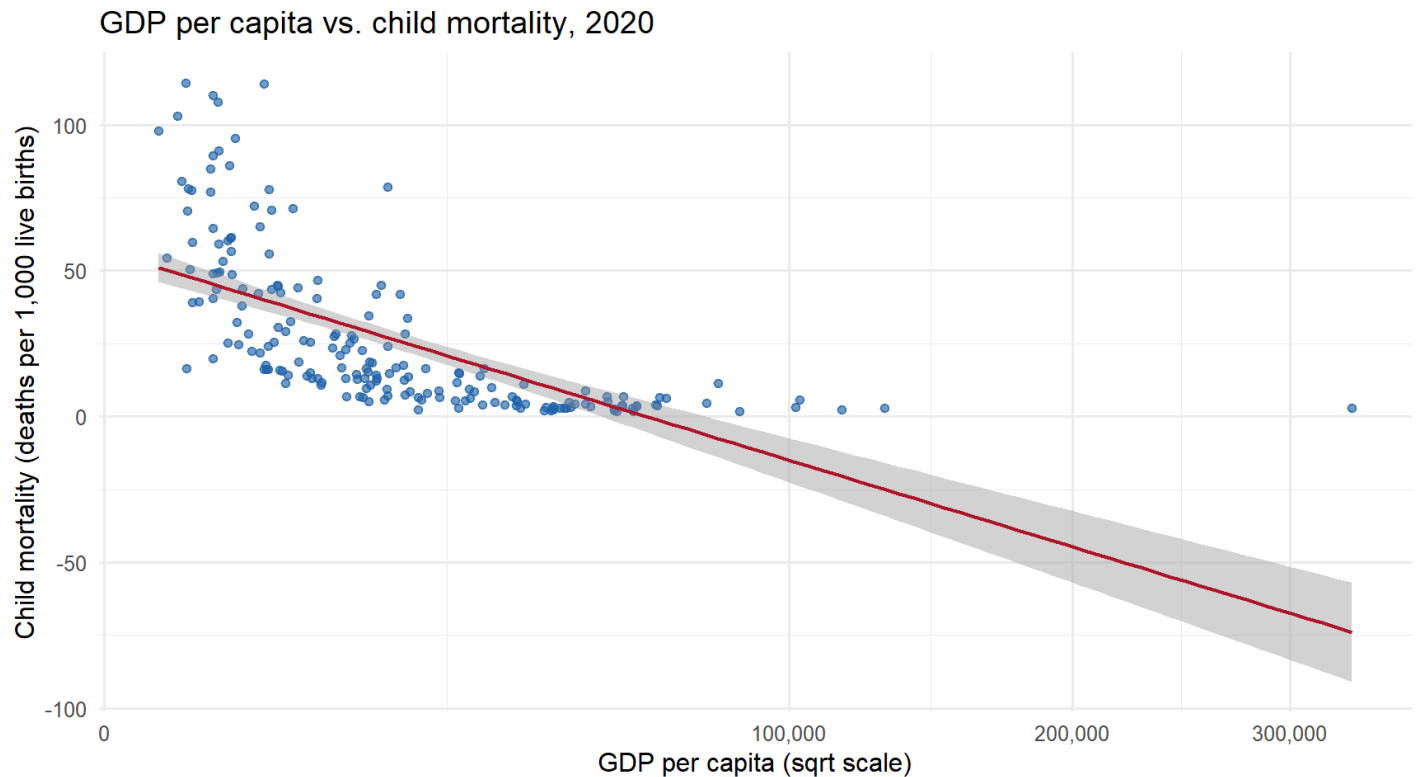


Child mortality in 2020

Child mortality (under age 5), 2020



Relationship between the two indicators



Interpretation

Countries with **higher GDP per capita in 2020 tend to have substantially lower under-five mortality**. Using a square-root scale for GDP (which spreads out low- and middle-income countries more than a linear axis), the association remains strongly negative (Pearson $r \approx -0.77$ between log GDP per capita and mortality among countries with both indicators). Wealthier economies—visible in high-GDP regions on the first map (North America, Western Europe, Oceania)—overlap with the lowest mortality rates on the second map. Conversely, many low-income countries in sub-Saharan Africa and parts of South Asia combine low GDP with mortality rates that remain far above 15.5 deaths per 1,000 births (the median among countries in this merged sample).

The pattern is not perfectly uniform: a few middle-income countries outperform or underperform relative to the trend line, reflecting differences in health systems, conflict, and inequality that GDP alone does not capture. Overall, however, **economic development and child survival move together at the global level in 2020**, consistent with better nutrition, sanitation, and health care where resources are greater.

Reflection

This was not a very easy assignment to do for me, as I had start-up issues setting up my Cursor app. This had to do with installing the extensions needed.

The first output I obtained had a full logarithmic scale for GDP per Capita, which I thought could be nicer. I asked to re-do this scale without further details. AI came up with a squareroot scale, which looked better.

The fact the correlation line extends into the negative for mortality rate seems a bit odd, and this actually should not be.

I learned a lot from this assignment, as many things can be taken care of by the AI.

The task for me to amend this text in the **report.md** was a bit difficult to find, as at first I was amending the text in the **report.html file**. However, this will be overwritten by a new knitting action.

End of Report